

Sign Language Interpretation Glove

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Abstract—This projects aims to make affordable and accessible sign interpretation glove. This by using gyroscope - accelerometer sensors attached to each finger, with a Raspberry pi mounted on the hand to act as the brain. With both gyroscope and accelerometer they could complement each others flaws with a complementary filter to calculate pitch and roll of the sensor. After two weeks code had been tested and given positive results on test data, but the sensors chosen had so small dimensions that they proved difficult to work with by hand. So to continue with this project different sensor or more precise lab equipment would be required to get any real time data acquisition.

I. INTRODUCTION

Sign language is a visible language expressed through hand gestures, eye gaze, facial expression and other small visible cues [?]. Today, sign language is spoken by more than 72 million people worldwide. However, the difficulty for those who use and rely on sign language to communicate with those who do not know it remains challenging, as is common in situations where individuals attempt to communicate across language barriers. Text-to-speech and transcription tools can partially bridge this gap but being unable to use their usual method of communication can be taxing. There are several commercially available experimental projects that demonstrate functional sign-to-speech gloves that aim to alleviate this problem.

A. Related Works

One notable example is BrightSign, which emphasizes a wide library of supported languages, customizable voice options, and a companion app that enhances the glove's usability at a steep price of 27031 sek at the time of writing[?].

Another significant development was presented by UCLA researchers in 2019, who created a prototype capable of recognizing American Sign Language (ASL) gestures, aiming for real-time translation into text and speech [?].

Although these projects demonstrate the feasibility of translating sign language through wearable technology, they have a high costs, expensive components or require specialized hardware and software ecosystems.

II. PROBLEM FORMULATION

A. Problem

Our project seeks to explore alternative approaches to make sign-to-speech translation more accessible and affordable. This would be done by using simple sensor technology called gyroscopes, and open source tools to achieve hand gesture recognition within the constraints of a rapid development

cycle. This has the potential to result in a cheap and more accessible sign interpretation glove and help those who have to rely on hand language communication.

B. Limitations

The whole project had some limitations, firstly a development time frame of two weeks and secondly a budget of 300 sek.

With limited time focus will be on getting sensor reading from a single finger. Once sensor reading was working for one finger, it can easily be replicated to the other.

Also, due to time limitation only pitch and roll will be considered to calculate orientation. Yaw is excluded, as an accelerometer can not measure yaw, and gyroscope only measures change of yaw and can not provide absolute orientation without additional equipment. So yaw is excluded because it requires additional components beyond the intended sensor setup.

Lastly, with a small budget the sensors had to be cheap. And while the LSM6DSO sensor is not ideal for the project, it is a reasonable choice given the constrained budget[?].

C. Research Question

- Are gyroscope-accelerometer sensors suitable sensors for use in a sign-to-speech glove?
- While being a cheaper option, will it provide accurate orientation and work reliably in real-time?

III. METHOD

A. Methodology

The LSM6DSO sensor integrates both a gyroscope and an accelerometer, allowing it to measure both angular velocity and linear acceleration [?]. The accelerometer has difficulties maintaining accuracy over short rapid movements, introducing noise to its measurement. And for the gyroscope, it tracks rotational motion but tends to drift due to the accumulation of small errors in each measurement. So to rely solely on one of the sensors makes it difficult to accurately estimate orientation.

To combat this flaw, a **Complementary filter** will be used to combine the strengths of both sensors[?]. These filters fuse the data, compensating for the weaknesses of each: the gyroscope drift is corrected by the accelerometer, and the accelerometer noise is smoothed by the gyroscope. This yields a more stable and accurate estimate of the sensor's orientation, which then

can be used to interpret specific hand positions or gestures used in sign language.

B. Hardware

A simple hardware setup was used, consisting of an LSM6DSO sensor connected to a Raspberry Pi Zero 2 W board [?]. This configuration enables real-time acquisition of inertial measurements, while being small and light enough to enable future integration into wearable form factors.

C. Software

The software was developed in Python as it is one of the most common programs used for raspberry pi coding [?]. And one of the main reason for it is due to it's large selection of libraries. One of them is the I2C protocol that will be used for for the communication between the sensor and Raspberry.

To enable parallel development of hardware and software, test data simulating sensor readings were first generated and used in place of live data. A simple Python code modeled a single finger starting in a reference position(fully extended), to being bent 45 degrees, then returns back to the reference position. With a known movement the measured value could be understood better and conclusions drawn about the Complementary filters performance could be evaluated. The simulated data are output as comma-separated values in the order:"accel-x, accel-y, accel-z, gyro-x, gyro-y, gyro-z".

From these raw values, pitch and roll angles were calculated independently using the accelerometer and gyroscope. The accelerometer-derived angles were calculated using trigonometric functions, based on the ratio of acceleration in each axis, while the gyroscope-derived angles were obtained by integrating the angular velocity over time. Both pitch and roll are calculated using similar mathematical expressions, with only a change in which axes are involved. For clarity and to avoid redundancy, only the pitch formulas are presented below.

For the accelerometer-based estimation, the pitch angle is computed using the x -axis acceleration and the combined magnitude of the y - and z -axes. Roll follows the same structure but switches to using the y -axis in the numerator. See equation ??.

For the gyroscope-based estimation, pitch is calculated by integrating the angular velocity around the x -axis, while roll uses the angular velocity around the y -axis. See equation ??.

As mentioned in the methodology subsection, each part of the sensor has limitations that will be mitigated by implementing a Complementary filter. This filter combines both data sources by weighting them according to their respective reliability: the short-term precision of the gyroscope and the long-term stability of the accelerometer. The resulting output provides a more accurate and stable estimate of the sensor's orientation, specifically the pitch (vertical bending) and roll (side tilt) of the finger. See equation ??.

$$\theta_{\text{pitch, accel}} = \arctan \left(\frac{a_x}{\sqrt{a_y^2 + a_z^2}} \right) \quad (1)$$

$$\theta_{\text{pitch, gyro}} = \theta_{\text{filtered, pitch}} + \omega_x \cdot \Delta t \quad (2)$$

$$\theta_{\text{pitch, filt}} = \alpha \cdot (\theta_{\text{pitch, gyro}}) + (1 - \alpha) \cdot \theta_{\text{pitch, accel}} \quad (3)$$

Variable definitions:

- $\theta_{\text{pitch, accel}}$ – Estimated pitch angle from the accelerometer.
- a_x, a_y, a_z – Acceleration values along the respective axes.
- $\theta_{\text{pitch, gyro}}$ – Estimated pitch angle from the gyroscope.
- $\theta_{\text{previous, pitch}}$ – Previously calculated pitch angle.
- ω_x – Angular velocity around the x -axis (in rad/s).
- Δt – Time elapsed since the previous measurement (in seconds).
- α – Weight constant (0.98 was used, commonly between 0.9 and 0.98).

The code process can be summarized in the following steps:

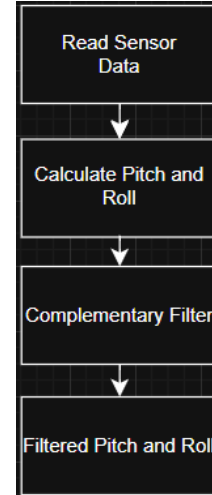


Figure 1. Overview of the software flowchart. The chart outlines the program's execution from simulated sensor input to orientation estimation using sensor fusion.

IV. RESULTS

Connection between Raspberry Pi and sensor were established but could not read because of an error due to faulty connections. The gyroscope with Complementary filter gave readings of the sensors pitch and roll rotation when used on the test data.

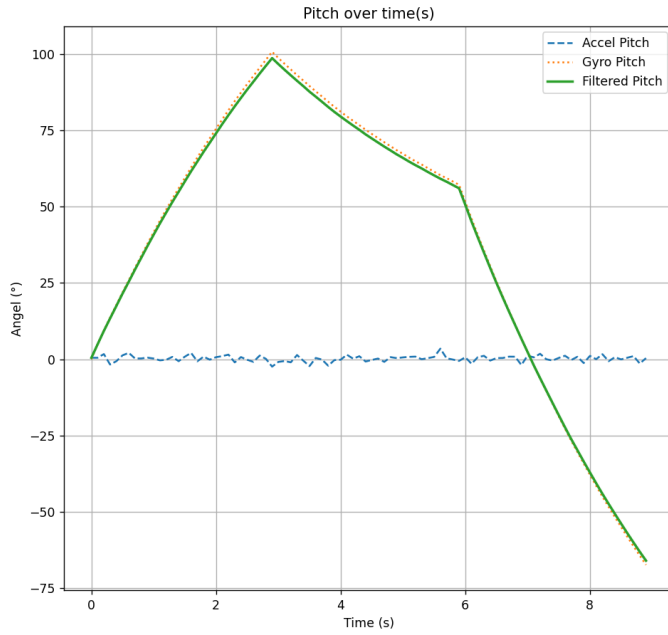


Figure 2. Theoretical result of the Pitch from using simulated data.

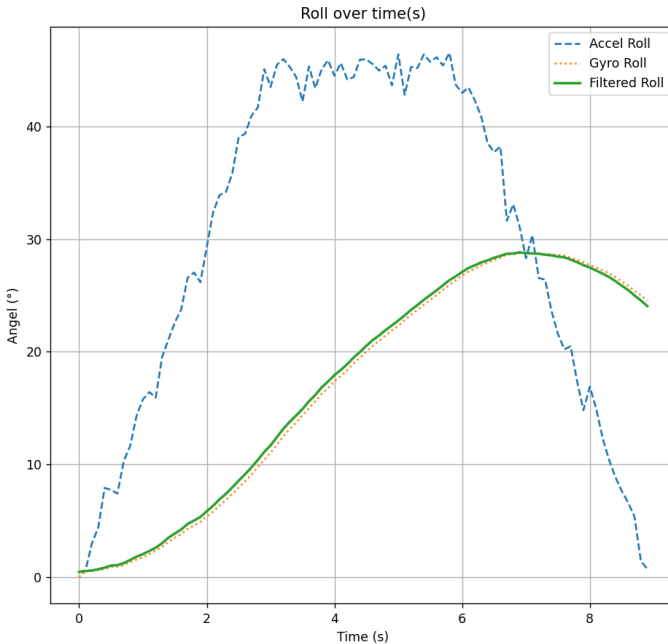


Figure 3. Theoretical result of the Roll from using simulated data.

V. DISCUSSION

The Raspberry Pi could detect the sensor but not read due to an I2C address connection. This was due to the small physical size of the sensor presented practical challenges during integration. With the sensor dimension of 2.5 x 3 x 0.83 mm and with limited laboratory equipment the group found it challenging, to manually solder connections with the required precision of 0.5 mm. This was something the group

did not account for, and is something that future work will have to solve, either with larger sensor or more precise lab equipment that can assist with the soldering process required precision.

Reflecting on the test data created, one can see from the graphs that the filtered plot follows the gyro scope but with a small correction. This gave the expected result as the weight constant was so large. The code worked for test data and could read and calculate Pitch and Roll, but because of the mentioned hardware struggles, it could not be tested in real time.

VI. CONCLUSION

Compared to more expensive alternatives such as flex sensors or optical systems, gyroscopes sensor shows potential as a low-cost option. With more suitable sensors, meaning larger and/or easier to work with, or a more precise lab equipment for the hardware would be required going forward.

With a cluster of sensors a detailed model of the user's hand position could be made and compared through simple case-based reasoning to output at least signed letters.

The software has only been evaluated using simulated sensor data. Therefore, no definitive conclusions can be drawn regarding its real-time performance or accuracy when applied to actual finger movements. Further testing with physical sensors and real-world motion would be required to validate the functionality and reliability of the system.

A. Future Work

For future work there are some key areas that can be developed.

With further development of the software the position of the sensor relative to a reference point could be estimated. With pitch, roll and known distance between sensor and joint axes, an approximate estimation in x,y and z can be inferred. This would need to take into account possible drift and noise, but this is based on the principle that a finger gesture corresponds to a unique combination of joint angles and direction. In other words, the sensor can achieve a unique combination of position and orientation for a given gesture.

Also, add the other fingers together with a mounting system on the hand, so that the entire hand can be monitored and measured. And for the LSM6DSO sensor, it only has two addresses so if it is going to be used for the entire hand, a multiplexer has to be used to be able to read all sensors.

Finally, it would be beneficial to work on a dataset of hand positions to have as reference and compare measurements to, and decide if specific signs are gestured.